Leo Villani

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Research Interests

I am interested in statistical theory and machine learning, with a focus on uncertainty, structure, and generalization in representations and graphs.

Education

Cornell University *Aug 2025 – Present*
Doctor of Philosophy in Statistics
GPA: 4.095 / 4.000

University of California, Berkeley *Aug 2021 – May 2025*
BA Applied Mathematics and Computer Science
Highest Honors (Mathematics), High Distinction (Computer Science)
GPA: 3.96 / 4.00

Papers and Manuscripts

Installing and Obstructing Heuristics: Learning Dynamics in Nim *Accepted, ICML 2026*
L. Villani, *S. Daniels, I. Yu, A. Sahai* *CompLearn Workshop*

The Geometry of Creativity: Semantic Null-Space Sampling for Generative Models *Under review*
AISTATS 2026
L. Villani, *P. Radchenko, B. Rava, D. Yang*

Confidence Sets for Causal Orderings with Interventional Data *Manuscript in preparation*
F. Fiocchi, ***L. Villani***, *Y. S. Wang*

Localization in Random Geometric Graphs from Sparse Anchors *Manuscript in preparation*
L. Villani, *Y. Wu, D. Yang*

Recovery and Isolation in Planted Hypergraph Matchings *Manuscript in preparation*
L. Villani, *D. Yang*

Experience

Quantitative Research Intern *Rowayton, CT*
Graham Capital Management *Jun 2026 – Aug 2026*

- Developing and calibrating a neural-network surrogate for a GARCH-style option-pricing model, using Monte Carlo simulations to generate option prices and implied volatilities across strikes and maturities.
- Evaluating alternative neural-network architectures and analyzing their tradeoffs in predictive accuracy, training stability, computational efficiency, and suitability for the option-pricing task.

Researcher *Berkeley, CA*
BLISS *Mar 2025 – Present*

- Built controlled Nim benchmarks and evaluated transformer shortcut learning, probing, and adversarial-invariance workflows on NCSA Delta.

Undergraduate Student Instructor (GSI) *Berkeley, CA*
UC Berkeley, Department of Mathematics *Jul 2023 – Jan 2025*

- Led discussion sections and office hours for mathematics courses; received the Outstanding GSI Award and a 6.7/7.0 student rating.

Technologies

Languages: Python, C++, C, R, SQL, Java, MATLAB, Go, Bash, LaTeX

Libraries: PyTorch, pandas, scikit-learn, statsmodels, Dask, Matplotlib

Tools: Git, Linux

Awards

Cornell Fellowship *Cornell University*
2025

ILR School Fellowship *Cornell ILR School*
Summer/Fall 2026

Outstanding Graduate Student Instructor (OGSI) Award *UC Berkeley*
Mar 2024